

Optimization & Machine Learning M

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Context and Objective

You have recently joined the Data Operations team at *Capital Bikeshare*, a public bicycle sharing system in Washington D.C. The company is facing a major logistical challenge: customer dissatisfaction due to empty stations (unable to rent a bike) and completely full stations (unable to return a bike).

Your manager has entrusted you with a two-part mission:

- **Machine Learning (Prediction):** Develop a model to predict the hourly demand for bike rentals based on temporal and meteorological factors.
- **Operations Research (Optimization) [Bonus]:** Use these predictions to optimize the overnight redistribution of bikes using logistics trucks.

Deliverables: A clean, commented, and well-structured Jupyter Notebook (`.ipynb`) containing your code and textual analyses.

Machine Learning Tasks

1. Data Exploration and Preprocessing (40%)

In the directory `data`, you are provided with two files: the dataset `bike_sharing_dataset.csv` and a text file `about_bike_sharing_dataset.txt` that describes the dataset's characteristics. Note that the target variable you need to predict is `cnt`.

The raw dataset may contain imperfections and anomalies. You must clean and prepare the dataset for model training.

- (a) **Exploration:** Load the dataset. Study the dimensions of the dataset, understand the meaning of each characteristic.
- (b) **Missing Values:** Identify and handle missing values logically. Pay attention to both scattered gaps and consecutive missing periods, as they may require different imputation strategies.
- (c) **Anomalies:** Identify physically impossible values and outliers. Correct, impute or delete them appropriately. *Hint: Verify that all values respect their documented ranges and that categorical variables only take valid values.*
- (d) **Feature Selection:** Identify and remove redundant columns, collinearities, and data leakage variables. Justify your choices. *Hint: You can generate a correlation matrix. Also inspect whether any column could directly reveal the target variable.*

- (e) **Feature Engineering:** Encode categorical and cyclic variables appropriately. *Hint: Apply sine/cosine transformations to all cyclic variables (e.g., hour of the day, month, day of the week). Apply one-hot encoding to nominal categorical variables.*
- (f) **Analysis:** Visualize the distribution of the target variable (`cnt`) and its relationship with temporal (hour, day, month) and meteorological variables. Comment.

2. Model Development and Evaluation (40%)

- (a) **Data Splitting:** Split the dataset into training (80%) and testing (20%) subsets.
- (b) **Baseline Model:** Train a baseline linear model (e.g., Linear Regression). Evaluate its performance using suitable metrics (e.g., RMSE, MAE, R-squared).
- (c) **Advanced Modeling:** Choose and train an appropriate non-linear machine learning algorithm (e.g., Random Forest).
- (d) **Fine-tuning:** Optimize your best model's hyperparameters using Grid Search to improve performance (e.g., explore the number of estimators, the learning rate, and the maximum tree depth).

3. Prediction and Interpretation (20%)

- (a) **Evaluation:** Use the trained model to make predictions on the testing data and assess its final performance.
 - (b) **Feature Importance:** Extract and visualize the most important features of your model. Do these align with your initial exploratory data analysis?
 - (c) **Conclusion:** Summarize your findings and provide business recommendations.
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Operations Research Task

4. Stock Optimization (Bonus)

Your predictions are meant to drive business decisions. Assume you isolate a sub-network of 20 stations. Comparing current stocks to the predicted demand for the next morning, you identify 8 stations with a surplus (supply) and 12 stations with a deficit (demand). In this scenario, the **total supply is insufficient to cover the total demand**: some deficit stations will inevitably remain partially or fully unserved.

You have been provided with an `or_data.json` file containing:

- a list of 20 stations;
- their current `supply_demand` status (positive integers for surplus, negative integers for deficit);
- a 20×20 symmetrical `distance_matrix` representing the travel costs in kilometers between all stations;
- a scalar `fixed_cost_per_route`: a fixed logistics cost (in km-equivalent) incurred each time a redistribution route between two stations is activated, regardless of how many bikes are sent;
- a scalar `penalty_per_missing_bike`: a penalty cost (in km-equivalent) for each bike that cannot be delivered to a deficit station.

Let $\mathcal{S}$ be the set of surplus stations and $\mathcal{D}$ the set of deficit stations. Let c_{ij} be the transportation cost (distance in km) between station $i \in \mathcal{S}$ and station $j \in \mathcal{D}$, f the fixed cost per activated route, and p the penalty per unserved bike. Define the following decision variables:

- $X_{ij} \in \mathbb{N}$: number of bikes moved from station $i \in \mathcal{S}$ to station $j \in \mathcal{D}$;
- $Y_{ij} \in \{0, 1\}$: equals 1 if route $(i \rightarrow j)$ is activated (i.e., at least one bike is sent), 0 otherwise;
- $Z_j \geq 0$: number of bikes left unserved at deficit station $j \in \mathcal{D}$.

(a) **Mathematical Formulation:** Write the Mixed Integer Linear Program (MILP) that minimizes the total cost, which combines: the proportional transportation cost c_{ij} per bike moved, the fixed activation cost f per route used, and the penalty p per unserved bike. Your formulation must ensure that:

- the number of bikes leaving each surplus station does not exceed its available supply;
- a route $(i \rightarrow j)$ can carry bikes only if it is activated ($Y_{ij} = 1$).

Hint: To link X_{ij} and Y_{ij} , use the big-M technique.

(b) **Implementation:** Import the JSON file and use the PuLP library in Python to solve this problem. PuLP is highly recommended because its intuitive, algebraic syntax makes modeling objective constraints straightforward (see the [Provide the optimal transportation plan](#) (the non-zero values of X_{ij}), the list of activated routes, the total number of unserved bikes per station, and the total optimized cost.